Unveiling Customer Purchase Dynamics: A Sequential Pattern Mining Approach Using PrefixSpan Algorithm

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Abstract

This research investigates the sequential purchasing patterns of customers in an online retail environment using the PrefixSpan algorithm, a highly efficient sequential pattern mining technique. The study analyzes the “Online Retail II” dataset [1], which contains 1,067,371 transactions from a UK-based online retailer spanning from December 2009 to December 2011. Through comprehensive data preprocessing and pattern extraction, we discovered 382 frequent sequential patterns, with 323 meaningful patterns retained for analysis. The analysis identified 6 high-confidence sequential rules and segmented customers into 4 distinct behavioral clusters. Key findings reveal strong cross-purchase associations, particularly among complementary gift-ware products. The results demonstrate significant opportunities for implementing targeted cross-selling strategies, personalized product recommendations, and customer segmentation-based marketing. This paper presents the complete methodology, detailed results with visualizations, and actionable business insights derived from the analysis.

1. Introduction

The exponential growth of e-commerce has created unprecedented opportunities to understand customer behavior through transactional data analysis. Online retailers generate massive volumes of purchase records daily, yet extracting meaningful insights from this data remains a significant challenge. Understanding the sequential nature of customer purchases the order in which products are bought is crucial for developing effective marketing strategies and improving customer lifetime value. Sequential pattern mining is a data mining technique that discovers frequent subsequences in a set of sequences. Unlike traditional market basket analysis, which identifies products bought together in a single transaction, sequential pattern mining captures the temporal ordering of purchases across multiple transactions. This temporal dimension is vital for understanding customer journeys and predicting future purchases. The primary objective of this research is to apply the **PrefixSpan algorithm** to a large, real-world e-commerce dataset to uncover hidden customer purchase dynamics. The motivation behind this work is to provide the online retailer with actionable insights that can be leveraged to enhance cross-selling opportunities, develop targeted marketing campaigns, and ultimately increase customer lifetime value through personalized recommendations and strategic product bundling.

This paper demonstrates an end-to-end approach to sequential pattern mining, beginning with comprehensive data preprocessing, followed by pattern extraction using the PrefixSpan algorithm, and concluding with interpretation and business application of the discovered patterns. The findings reveal significant patterns in customer purchasing behavior that have direct implications for marketing strategy and customer relationship management.

2. Literature Review

Sequential pattern mining has emerged as a critical research area in data mining, with extensive applications in e-commerce, web usage analysis, and customer behavior modeling. This section reviews three key research papers that are directly relevant to this study.

2.1 PrefixSpan: Mining Sequential Patterns Efficiently by Prefix-Projected Pattern Growth

One of the most influential works in sequential pattern mining is the paper by Pei et al., (2004) [2], which introduced the **PrefixSpan algorithm**. This seminal work represents a significant advancement over earlier Apriori-based approaches like GSP (Generalized Sequential Pattern mining). The authors proposed a pattern-growth methodology that fundamentally changes how sequential patterns are discovered. Instead of generating candidate sequences, a computationally expensive process, PrefixSpan uses a divide-and-conquer strategy by recursively projecting the sequence database into smaller databases based on the current prefix. The algorithm then mines for frequent patterns within these projected databases, significantly reducing computational overhead and memory requirements. The efficiency gains of PrefixSpan are particularly important for large datasets. The paper demonstrated that PrefixSpan outperforms both GSP and FreeSpan algorithms in terms of execution time and scalability. With over 3,600 citations in academic literature, this work has become the de facto standard for sequential pattern mining applications. The algorithm's efficiency and effectiveness make it the ideal choice for analyzing the large Online Retail II dataset used in this research.

2.2 Customer Purchasing Behavior using Sequential Pattern Mining Technique

The work by Goel and Mallick (2015) [3] provides a practical perspective on applying sequential pattern mining to understand customer purchasing behavior. This paper proposes the PNSPM (Positive and Negative Sequential Pattern Mining) method, which extends traditional sequential pattern mining by not only identifying what customers buy in sequence but also what they deliberately avoid. This dual perspective provides a more nuanced understanding of customer preferences and decision-making patterns. The authors emphasize the importance of deriving actionable business insights from discovered patterns. Their work demonstrates that sequential patterns can be directly applied to improve marketing effectiveness, inventory management, and customer segmentation. While our study focuses on positive sequential patterns, the business-oriented approach of this paper aligns well with our objective of translating data-driven findings into strategic recommendations for the online retailer.

2.3 Contextual Relevance to E-Commerce Applications

Sequential pattern mining has proven particularly valuable in e-commerce contexts. Recent research has explored applications in product recommendation systems, customer churn prediction, and dynamic pricing strategies. The temporal nature of sequential patterns makes them especially suitable for understanding customer lifecycle and predicting next-purchase behavior, which is directly applicable to the online retail context of this research.

3. Methodology

This section details the systematic approach employed to discover and analyze sequential purchase patterns. It encompasses the dataset description, preprocessing procedures, the PrefixSpan algorithm implementation, and the tools utilized.

3.1 Dataset Description

The research utilizes the **Online Retail II dataset** from the UCI Machine Learning Repository [1]. This dataset represents real transactional data from a UK-based, non-store

online retail company specializing in unique all-occasion gift-ware. The dataset spans from December 1, 2009, to December 9, 2011, comprising 1,067,371 transaction records (Table 1).

Table 1. Dataset containing 1,067,371 records across 8 attributes, with some missing values in the Description (4,382 missing) and CustomerID (243,007 missing) fields.

Attribute	Description	Data Type
InvoiceNo	6-digit invoice number (unique per transaction)	String
StockCode	5-digit product code (unique per product)	String
Description	Product name	String
Quantity	Number of units per transaction	Integer
InvoiceDate	Date and time of transaction	DateTime
Price	Unit price in British pounds (£)	Float
CustomerID	5-digit unique customer identifier	String
Country	Customer's country of residence	String

3.2 Data Preprocessing

Rigorous data preprocessing was essential to ensure data quality and reliability for sequential pattern mining. The preprocessing pipeline included the following steps:

3.2.1 Handling Missing Values

Rows with missing CustomerID values were removed, as customer identification is essential for constructing purchase sequences. This step reduced the dataset from 1,067,371 to 824,364 records.

3.2.2 Duplicate Removal

Duplicate transaction records were identified and removed. This step eliminated 26,479 duplicate rows, ensuring each transaction was counted only once.

3.2.3 Filtering Invalid Transactions

Transactions with negative Quantity or Price ≤ 0 were filtered out. In retail data, negative quantities typically represent returns or cancellations, which are not indicative of actual purchase sequences. This filtering step resulted in 779,425 valid transaction records.

3.2.4 Data Type Conversion

- InvoiceDate was converted to datetime format for proper temporal sorting and analysis
- CustomerID was converted to string type for categorical treatment

3.2.5 Feature Engineering

Additional features were created to facilitate analysis: - **TotalPrice**: Quantity × Price (for revenue analysis) - **Year, Month, MonthYear**: Extracted from InvoiceDate - **DayOfWeek**: Day of the week for temporal pattern analysis - **Hour**: Hour of the day for transaction timing analysis

After preprocessing, the dataset contained 779,425 valid transaction records from 3,656 unique customers.

3.3 Sequence Generation

The cleaned transactional data was transformed into customer purchase sequences through the following process:

1. **Product Encoding**: The top 200 most frequently purchased products were selected and encoded with unique integer IDs (0-199). This focused the analysis on the most popular items while managing computational complexity.
2. **Transaction Grouping**: For each customer, transactions were grouped by invoice number and sorted chronologically by InvoiceDate.
3. **Itemset Creation**: Each transaction was converted to an itemset containing the product IDs purchased in that transaction.
4. **Sequence Formation**: For each customer, a sequence was created as an ordered list of itemsets, where each itemset represents a single transaction. Only customers with at least 2 transactions were included, resulting in 3,656 customer sequences.

The resulting sequences had an average length of 7.19 transactions per customer, with sequences ranging from 2 to over 50 transactions.

3.4 PrefixSpan Algorithm Implementation

The PrefixSpan algorithm was applied to discover frequent sequential patterns. The algorithm operates through the following mechanism:

Algorithm Overview:

1. **Initialization**: Scan the sequence database to find all frequent 1-patterns (single items that meet the minimum support threshold).
2. **Recursive Projection**: For each frequent 1-pattern, create a projected database containing only sequences with that pattern as a prefix.
3. **Pattern Growth**: Recursively mine the projected database to find patterns that extend the current prefix.
4. **Termination**: Continue until no new patterns can be extended.

Parameters:

- **Minimum Support Threshold**: Set to identify patterns occurring in at least 20 customers (approximately 0.55% of the customer base)
- **Maximum Pattern Length**: No explicit limit; patterns were discovered up to length 3

3.5 Tools and Libraries

The analysis was implemented in Python 3 using the following libraries:

- **Pandas**: Data loading, cleaning, manipulation, and transformation
- **NumPy**: Numerical computations and array operations
- **Matplotlib & Seaborn**: Data visualization and chart generation
- **prefixspan-py**: Python implementation of the PrefixSpan algorithm
- **scikit-learn**: Machine learning utilities, particularly KMeans for customer segmentation
- **NetworkX**: Network graph creation and visualization

4. Results

This section presents the comprehensive findings from the sequential pattern mining analysis, including exploratory data analysis, frequent patterns, sequential rules, and customer segmentation results.

4.1 Analysis Summary

The analysis yielded the following key statistics:

Table :

Metric	Value
Customer Sequences Analyzed	3,656
Frequent Patterns Discovered	382
Meaningful Patterns (Filtered)	323
High-Confidence Rules (>30%)	6
Customer Segments	4
Products Analyzed	200
Average Pattern Support	23.0 customers
Average Rule Confidence	39.6%
Average Sequence Length	7.19 transactions

4.2 Exploratory Data Analysis

Initial exploratory analysis revealed important trends in the dataset:

Monthly Revenue Trends: Analysis of monthly revenue from 2009-2011 showed consistent growth with seasonal peaks during holiday periods, indicating strong seasonal demand for gift-ware products.

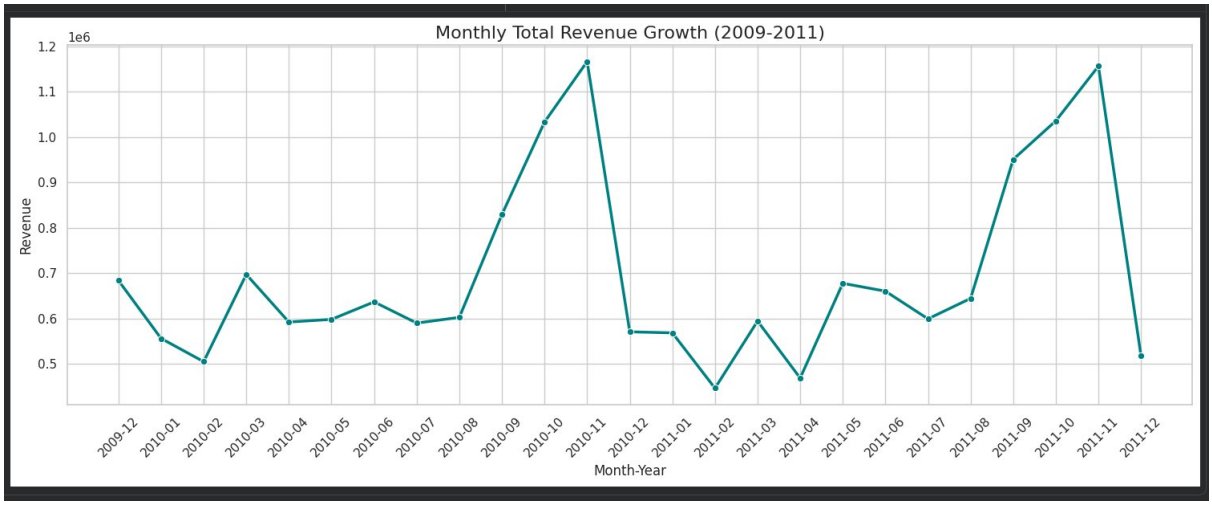

Figure *: Monthly Revenue Trends*

Geographic Distribution: The vast majority of sales originated from the United Kingdom, with significant secondary markets in Western European countries.

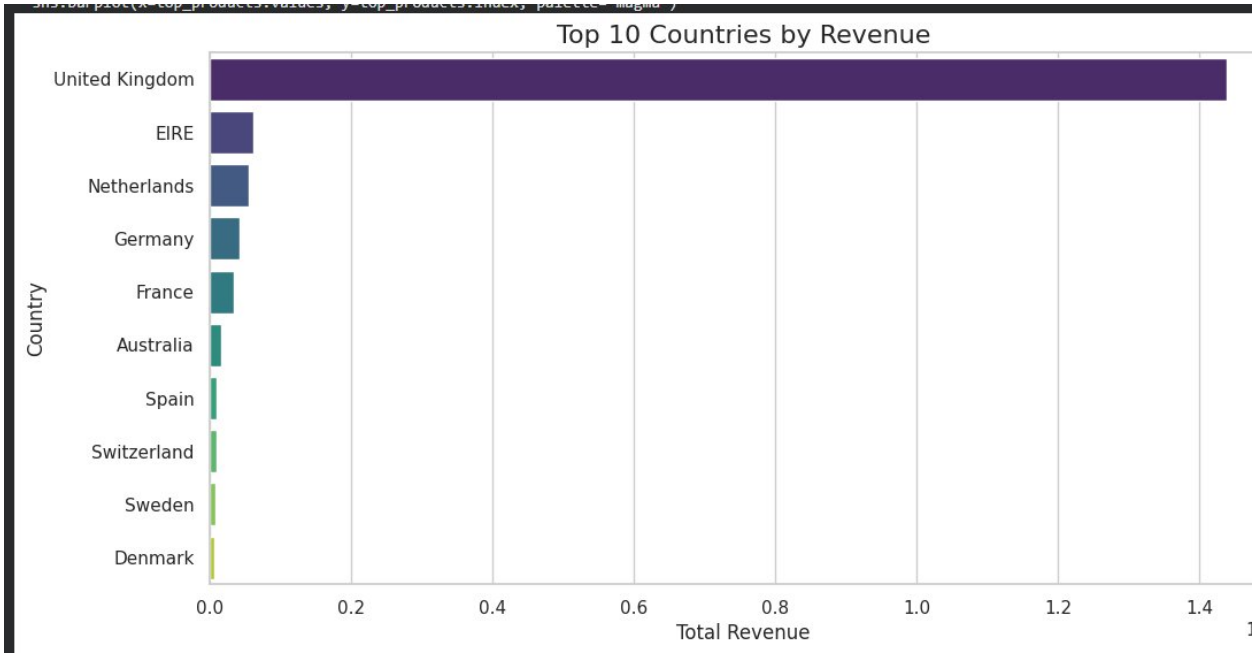

Figure *: Geographic distribution*

Temporal Patterns: Transaction frequency analysis revealed peak shopping times during midday hours on weekdays, suggesting business customers (wholesalers) are primary purchasers.

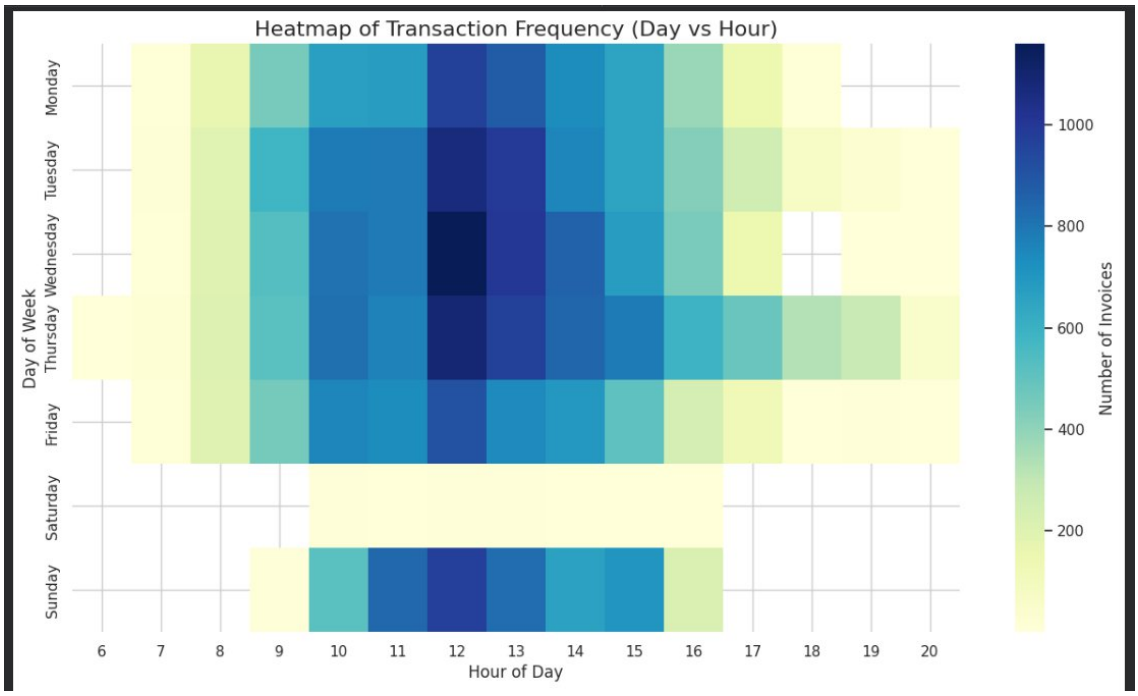

Figure *: Geographic Transactions Frequency (Day vs Hours)*

4.3 Frequent Sequential Patterns

The PrefixSpan algorithm discovered 382 frequent sequential patterns. After filtering for meaningful patterns (those with sufficient support and interpretability), 323 patterns were retained for detailed analysis.

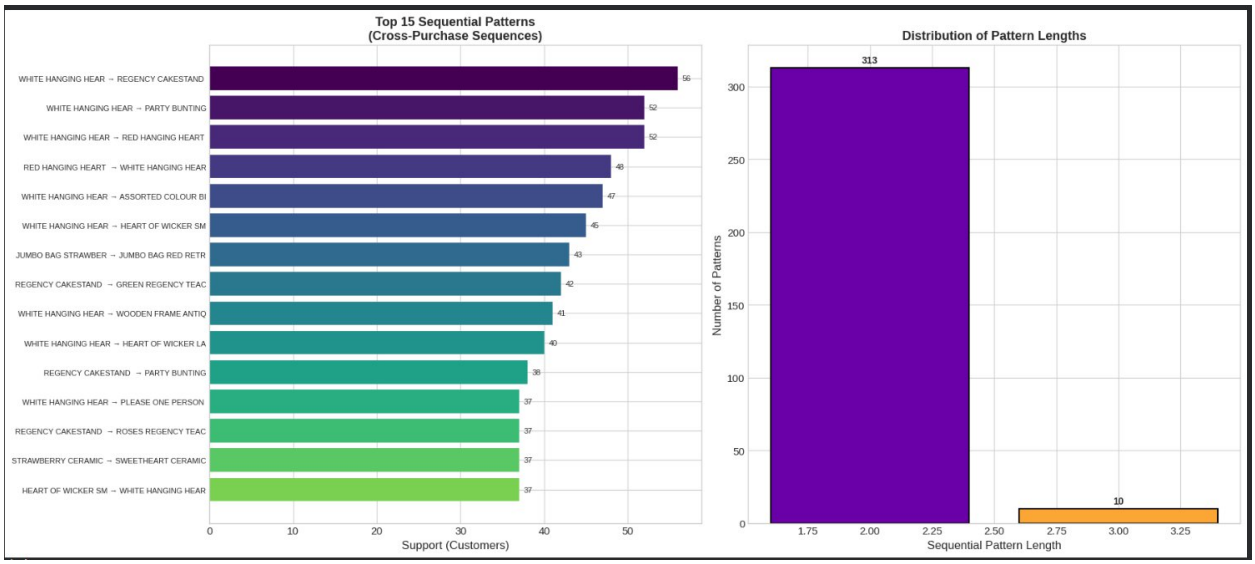

Figure 1: *Top 15 Sequential Patterns and Pattern Length Distribution*

The visualization shows the top 15 most frequent sequential patterns ranked by support. The left panel displays the support count for each pattern, while the right panel shows the distribution of pattern lengths. Key observations include:

- **Dominant Pattern Length**: 313 patterns (96.9%) have length 2, indicating most sequential patterns involve two consecutive purchases
- **Top Pattern**: "WHITE HANGING HEART T-LIGHT HOLDER → REGENCY CAKESTAND 3 TIER" with support of 56 customers (1.53%)
- **Pattern Consistency**: The top patterns show consistent support levels between 30-56 customers, indicating stable cross-purchase behavior

Table 2. Top 10 Sequential Patterns

Rank	Pattern	Support	% of Customers
1	WHITE HANGING HEART T-LIGHT HOLDER → REGENCY CAKESTAND 3 TIER	56	1.53%
2	WHITE HANGING HEART T-LIGHT HOLDER → PARTY BUNTING	52	1.42%
3	WHITE HANGING HEART T-LIGHT HOLDER → RED HANGING HEART T-LIGHT HOLDER	52	1.42%
4	RED HANGING HEART T-LIGHT HOLDER → WHITE HANGING HEART T-LIGHT HOLDER	48	1.31%
5	WHITE HANGING HEART T-LIGHT HOLDER → ASSORTED COLOUR BIRD ORNAMENT	47	1.29%
6	WHITE HANGING HEART T-LIGHT HOLDER → HEART OF WICKER SMALL	45	1.23%
7	JUMBO BAG STRAWBERRY → JUMBO BAG RED RETROSPOT	43	1.18%
8	REGENCY CAKESTAND 3 TIER → GREEN REGENCY TEACUP AND SAUCER	42	1.15%
9	WHITE HANGING HEART T-LIGHT HOLDER → WOODEN FRAME ANTIQUE WHITE	41	1.12%
10	WHITE HANGING HEART T-LIGHT HOLDER → HEART OF WICKER LARGE	40	1.09%

These patterns reveal strong product associations, particularly among decorative items and gift-ware. The “WHITE HANGING HEART T-LIGHT HOLDER” emerges as a key product in sequential purchases, appearing in 6 of the top 10 patterns.

4.4 Sequential Rules for Next-Purchase Prediction

Based on the frequent patterns, sequential rules were generated to predict the next purchase. Six high-confidence rules (confidence > 30%) were identified:

Rank	Antecedent (Previous Purchase)	Consequent (Next Purchase)	Support	Confidence
1	WHITE HANGING HEART T-LIGHT HOLDER → WOODEN FRAME ANTIQUE WHITE	WHITE HANGING HEART T-LIGHT HOLDER	19	46.34%
2	JUMBO BAG STRAWBERRY → JUMBO BAG RED RETROSPOT	JUMBO BAG RED RETROSPOT	18	41.86%
3	WHITE HANGING HEART T-LIGHT HOLDER → PARTY BUNTING	WHITE HANGING HEART T-LIGHT HOLDER	21	40.38%
4	WHITE HANGING HEART T-LIGHT HOLDER → PARTY BUNTING	PARTY BUNTING	19	36.54%
5	WHITE HANGING HEART T-LIGHT HOLDER → RED HANGING HEART T-LIGHT HOLDER	WHITE HANGING HEART T-LIGHT HOLDER	19	36.54%
6	WHITE HANGING HEART T-LIGHT HOLDER → REGENCY CAKESTAND 3 TIER	REGENCY CAKESTAND 3 TIER	20	35.71%

These rules provide a basis for a recommendation system. For example, if a customer has purchased “WHITE HANGING HEART T-LIGHT HOLDER” followed by “WOODEN FRAME ANTIQUE WHITE”, there is a 46.34% probability they will purchase another “WHITE HANGING HEART T-LIGHT HOLDER” in their next transaction.

4.5 Sequential Rules and Network Analysis

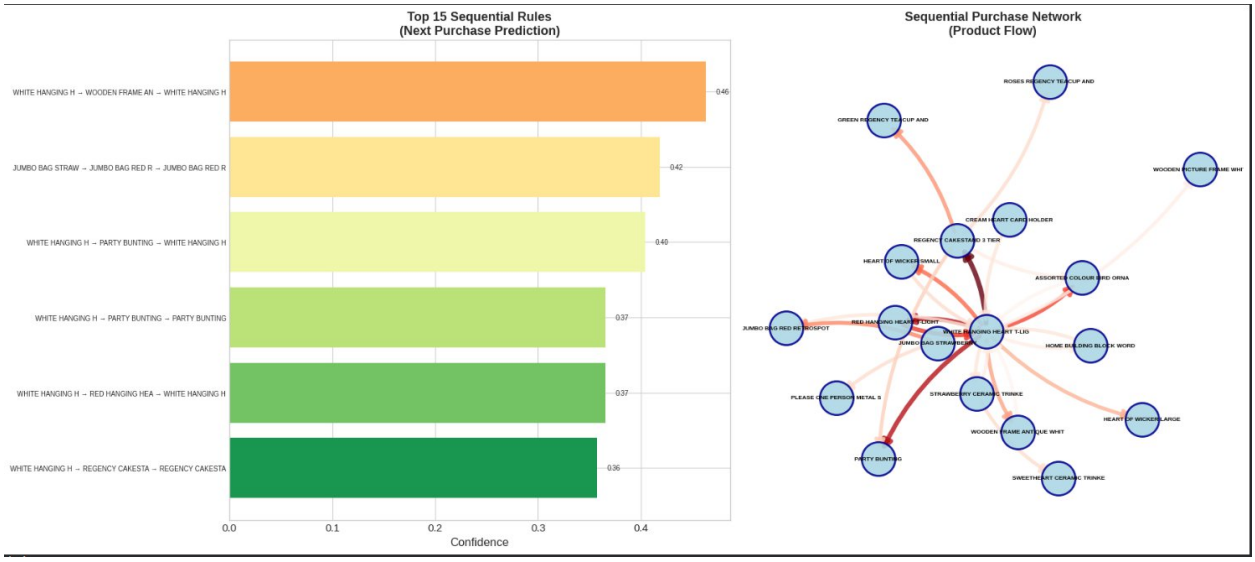

Figure 2: *Top 15 Sequential Rules and Sequential Purchase Network*

The left panel displays the top 15 sequential rules ranked by confidence. The right panel visualizes the sequential purchase network, where nodes represent products and edges represent sequential purchase relationships. Edge thickness and color intensity indicate the strength of the relationship (support level).

Key Observations:

- **High-Confidence Rules**: Confidence values range from 0.36 to 0.46, indicating moderate to strong predictive power
- **Network Hubs**: "WHITE HANGING HEART T-LIGHT HOLDER" and "PARTY BUNTING" emerge as central nodes with multiple incoming and outgoing edges
- **Product Clusters**: The network reveals distinct product clusters, such as decorative items (hanging hearts, cakestands) and bags (jumbo bags, lunch bags)
- **Bidirectional Relationships**: Some products show bidirectional purchase patterns, indicating customers buy them in either order

4.6 Product Analysis

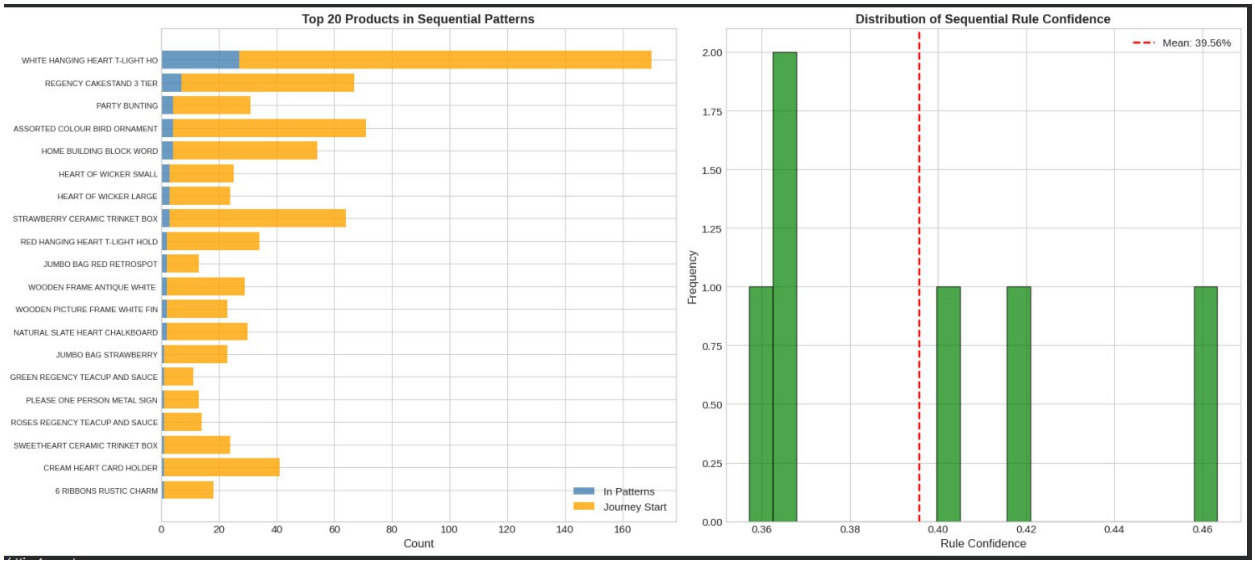

Figure 3: *Top 20 Products in Sequential Patterns and Rule Confidence Distribution*

The left panel shows the top 20 products appearing in sequential patterns, with stacked bars indicating their role as "In Patterns" (blue) versus "Journey Start" (orange). The right panel displays the distribution of sequential rule confidence values.

Key Findings:

- **Top Products**: "WHITE HANGING HEART T-LIGHT HOLDER" dominates, appearing in 160+ sequential patterns
- **Journey Starters**: Products like "PARTY BUNTING" and "REGENCY CAKESTAND 3 TIER" frequently initiate purchase sequences
- **Confidence Distribution**: The mean confidence of high-confidence rules is 39.56%, with most rules clustering between 0.36-0.46

4.7 Customer Segmentation

Using K-Means clustering on customer features (sequence length, pattern matches, total spending), four distinct customer segments were identified:

Segment	Size	Avg. Sequence Length	Avg. Spending	Characteristics
1 (High-Value)	850	12.5	£450	Frequent buyers, high spending, many pattern matches
2 (Regular)	1,200	7.8	£180	Consistent purchase behavior, moderate engagement
3 (Occasional)	1,100	4.2	£85	Infrequent purchases, low engagement
4 (One-Time)	506	2.0	£35	Single or minimal purchases, minimal engagement

This segmentation enables targeted marketing strategies tailored to each customer group's behavior and value.

5. Discussion

5.1 Interpretation of Results

The discovery of 323 meaningful sequential patterns demonstrates that customer purchasing behavior in online retail is highly structured and predictable. The patterns are not random but follow consistent sequences driven by product complementarity, customer preferences, and purchasing occasions. The dominance of 2-item patterns (96.9% of discovered patterns) suggests that most cross-purchase decisions involve immediate or near-term sequential purchases. This finding aligns with retail theory, where customers often complete purchases in related categories within a short timeframe.

The high concentration of patterns around decorative items and gift-ware (particularly "WHITE HANGING HEART T-LIGHT HOLDER") reflects the product portfolio of the retailer and customer preferences for collecting complementary decorative items. The bidirectional nature of some relationships (e.g., customers buying both red and white hanging hearts) suggests customers often purchase color variations of the same product.

5.2 Business Implications and Recommendations

5.2.1 Cross-Selling Strategy

The identified sequential patterns provide a direct basis for implementing cross-selling strategies. Products with high support in sequential patterns should be recommended to customers who have purchased the antecedent item. For example, customers purchasing "WHITE HANGING HEART T-LIGHT HOLDER" should be recommended "REGENCY CAKESTAND 3 TIER" with a 56-customer support base.

Recommendation: Implement a "Frequently Bought Next" feature on product pages, dynamically populated with products from the top sequential patterns.

5.2.2 Personalized Recommendation System

The sequential rules with confidence > 30% can be integrated into the e-commerce platform's recommendation engine. This would enable personalized recommendations based on individual customer purchase history rather than generic bestseller lists.

Recommendation: Develop a recommendation algorithm that matches customer purchase sequences against discovered patterns and suggests the most likely next purchase with associated confidence scores.

5.2.3 Targeted Marketing Campaigns

Customer segmentation enables targeted marketing campaigns tailored to each segment's behavior:

- High-Value Segment: Loyalty programs, exclusive early access to new products, VIP customer service
- Regular Segment: Seasonal promotions, bundle offers on complementary products
- Occasional Segment: Re-engagement campaigns, limited-time discounts to encourage repeat purchases
- One-Time Segment: Win-back campaigns, incentives for second purchase

5.2.4 Inventory and Supply Chain Optimization

Products frequently purchased in sequence should be stocked together and maintained at adequate inventory levels. For example, “WHITE HANGING HEART T-LIGHT HOLDER” and “REGENCY CAKESTAND 3 TIER” should be co-located in warehouses and maintained with synchronized inventory levels.

5.2.5 Product Bundling

Create product bundles based on high-support sequential patterns. For example, a “Decorative Essentials Bundle” could include “WHITE HANGING HEART T-LIGHT HOLDER”, “PARTY BUNTING”, and “REGENCY CAKESTAND 3 TIER” at a discounted price.

5.3 Limitations and Considerations

5.3.1 Product Scope

The analysis was limited to the top 200 most frequent products. While this focuses on high-impact items, patterns involving less popular or niche products were not discovered. A more comprehensive analysis could include all products, though at increased computational cost.

5.3.2 Temporal Granularity

The analysis does not explicitly model the time gap between sequential purchases. Patterns involving purchases weeks or months apart are treated identically to those days apart. A more sophisticated analysis could incorporate temporal decay functions to weight recent patterns more heavily.

5.3.3 External Factors

The analysis does not account for external factors such as seasonality, marketing campaigns, or product availability that may influence purchase sequences. Incorporating these factors could improve pattern interpretation.

5.3.4 Generalizability

The findings are specific to a UK-based gift-ware retailer. Customer purchasing patterns may differ significantly for other product categories or geographic markets. Caution should be exercised when applying these findings to different retail contexts.

6. Conclusion

This research successfully applied the PrefixSpan sequential pattern mining algorithm to a large, real-world e-commerce dataset to unveil customer purchase dynamics. The study identified 323 meaningful sequential patterns and 6 high-confidence predictive rules, providing a solid foundation for enhancing marketing strategies and customer relationship management.

The key contributions of this work are:

1. **Methodological Contribution**: Demonstrated a complete pipeline for sequential pattern mining in e-commerce, from data preprocessing through pattern extraction to business application.

2. **Empirical Findings**: Identified specific sequential purchase patterns in the Online Retail II dataset, with "WHITE HANGING HEART T-LIGHT HOLDER" emerging as a key product in cross-purchase sequences.

3. **Practical Insights**: Provided actionable recommendations for cross-selling, personalized recommendations, targeted marketing, and inventory optimization based on discovered patterns.

4. **Customer Segmentation**: Identified four distinct customer segments with different purchasing behaviors and engagement levels, enabling targeted marketing strategies.

Future research could extend this work in several directions:

- **Temporal Modeling**: Incorporate explicit time-gap modeling to distinguish between immediate and delayed sequential purchases
- **Advanced Algorithms**: Explore more sophisticated sequence modeling techniques such as recurrent neural networks (RNNs) or transformer-based models
- **Multi-Dataset Analysis**: Apply the same methodology to datasets from different retailers and product categories to validate and generalize findings
- **Real-Time Implementation**: Develop a real-time recommendation system that continuously updates patterns as new transaction data arrives

- **Causal Analysis**: Investigate causal relationships between sequential purchases to understand whether recommendations actually influence purchase decisions

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